



Arab Academy for Science, Technology & Maritime Transport College of Computing and Information Technology

Course: System modelling and Simulation

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Section # 3

Problem 1

A news boy buys paper for 15P.T.each and sells them for 25P.T.each unsold papers have no value in analyzing past sales it is found that three demand distributions exist, depending on news events in the papers. A good interesting news day yields to many potential customers, typical and poor days yield to less interest. Historically 20% of days have been "good", 50% "typical", 30% "Poor", The demand distributions are summarized :

Types of new days

Demand	Good 00-->19	Typical 20-->69	Poor 70--> 99
40	0.05 00 --> 04		
50	0.2 05 --> 24	0.03 00 -->02	
60	0.41 25 --> 65	0.07 03 -->09	
70	0.26 66 --> 91	0.28 10 -->37	
80	0.08 92-->99	0.3 38-->67	0.02 00 --> 01
90		0.25 68-->92	0.15 02 --> 16
100		0.07 93-->99	0.28 17 --> 44
110			0.24 45 --> 68
120			0.12 69 --> 80
130			0.09 81 --> 89
140			0.06 90 --> 95
150			0.04 96 --> 99

Simulate 10 days period and determine mean daily profit, max profit, min profit if he decides to buy 80 papers each day. Use the following R.N :-

09 65 69 66 57 64 84 24 92 22 49 50 61 01 45 48 48 45 52 35

Answer

Note : We have two digit random numbers that means that range of numbers 00 → 99

Good take numbers from 00 → 19 , Typical take numbers from 20 → 69 Poor take numbers from 70 → 99, 80 paper each day → Max buy 80 paper per day

Note: we take 2 Random number first one to determine the type of new day , The second one to determine demand " how much he sell in that day ".

1st Day :

RN = 09 → Good

RN = 65 → demand = 60

Profit = $60 \times 25 - 80 \times 15 = 300$ P.T

2nd Day :

RN= 69 → Typical

RN = 66 → demand = 80

Profit = $80 \times 25 - 80 \times 15 = 800$ P.T

3rd Day

RN = 57 → Typical

RN = 64 → demand = 80

Profit = $80 \times 25 - 80 \times 15 = 800$ P.T

4th Day :

RN= 84 → poor

RN = 24 → demand = 100 = ~~80~~ → because max of buy is 80 paper each day

Profit = $80 \times 25 - 80 \times 15 = 800$ P.T ; we calculate with demand 80 NOT 100.

5th Day :

RN= 92 → poor

RN = 22 → demand = 100 = ~~80~~ → because max of buy is 80 paper each day

Profit = $80 \times 25 - 80 \times 15 = 800$ P.T ; we calculate with demand 80 NOT 100.

6th Day :

RN= 49 → Typical RN = 50 →

demand = 80 Profit = $80 \times 25 -$

$80 \times 15 = 800$ P.T

7th Day :

RN= 61 → Typical

RN= 01 → demand = 50

Profit = $50 \times 25 - 80 \times 15 = 50$ P.T

8th Day :

RN = 45 → Typical

RN= 48 → demand = 80

Profit = $80 \times 25 - 80 \times 15 = 800$ P.T

9th Day :

RN = 48 → Typical

RN= 45 → demand = 80

Profit = $80 \times 25 - 80 \times 15 = 800$ P.T

10th Day :

RN= 52 → Typical

RN= 35 → demand = 70

Profit = $70 \times 25 - 80 \times 15 = 550$ P.T

- MIN profit we found it in 7th day which is **50 P.T**
- MAX profit is **800 P.T**
- Mean daily profit $300+800+800+800+800+800+50+800+800+550/10=$ **650 PT**